

Machine Learning Homework 3

3.1 Give decision trees to represent the following boolean functions:

(a) $A \wedge \neg B$

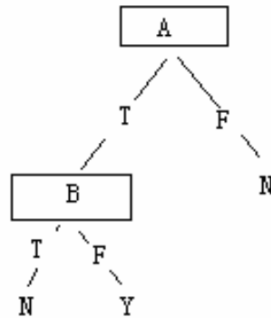
(b) $A \vee [B \wedge C]$

(c) $A \text{ XOR } B$

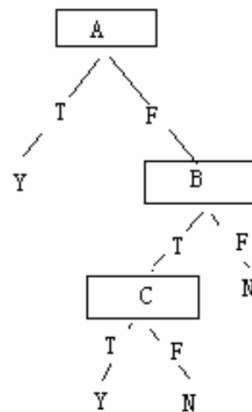
(d) $[A \wedge B] \vee [C \wedge D]$

Answer:

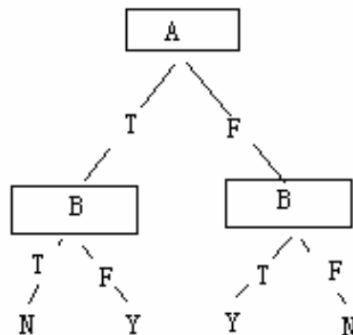
(a) $A \wedge \neg B$



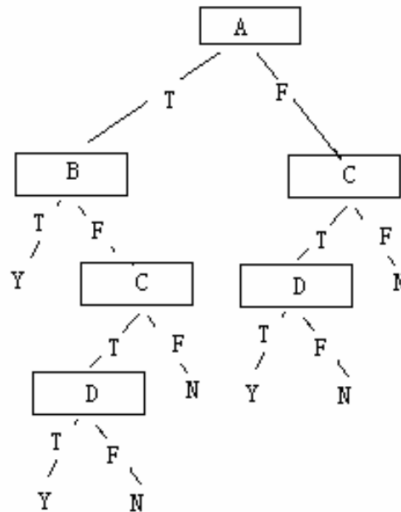
(b) $A \vee [B \wedge C]$



(c) $A \text{ XOR } B$



(d) $[A \wedge B] \vee [C \wedge D]$



3.2. Consider the following set of training examples:

Instance	Classification	a_1	a_2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

(a) What is the entropy of this collection of training examples with respect to the target function classification?

Answer:

$$\begin{aligned}
 Entropy(S) &= -p_{+} \log_2 p_{+} - p_{-} \log_2 p_{-} \\
 &= -(3/6) \log_2 (3/6) - (3/6) \log_2 (3/6) \\
 &= 1
 \end{aligned}$$

(b) What is the information gain of a_2 relative to these training examples?

Answer:

$$\begin{aligned}
 Gain(S, a_2) &= Entropy(S) - \sum_{v \in Value(a_2)} \frac{S_v}{S} Entropy(S_v) \\
 &= 1 - (4/6) \left[-(2/4) \log_2 (2/4) - (2/4) \log_2 (2/4) \right] - (2/6) \left[-(1/2) \log_2 (1/2) - (1/2) \log_2 (1/2) \right] \\
 &= 0
 \end{aligned}$$

3.3. True or false: If decision tree D2 is an elaboration of tree D1, then D1 is

more-general-than D2. Assume D1 and D2 are decision trees representing arbitrary Boolean functions, and that D2 is an elaboration of D1 if ID3 could extend D1 into D2. If true, give a proof; if false, a counterexample. (More-general-than is defined in Chapter 2.)

Answer:

False.

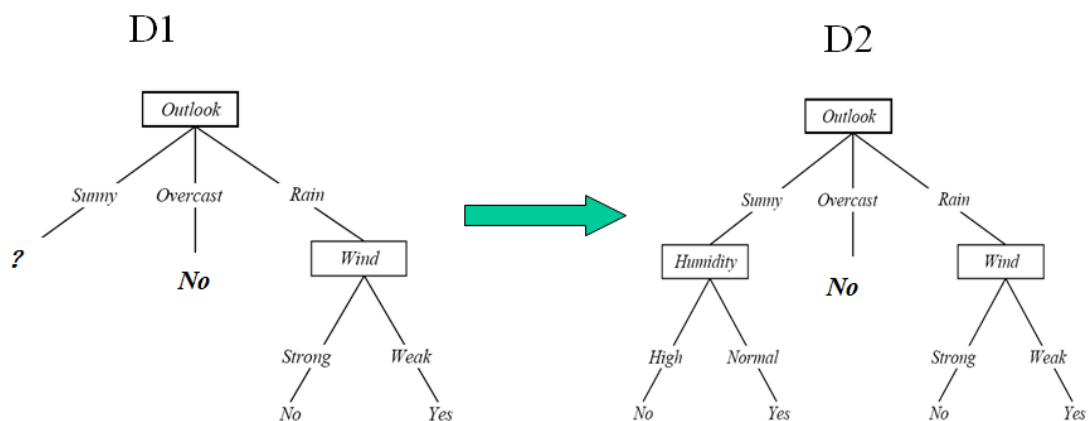
Proof:

According to the definition of more_general_than:

Definition: Let h_j and h_k be boolean-valued functions defined over X . Then h_j is **more_general_than_or_equal_to** h_k (written $h_j \succeq_g h_k$) if and only if

$$(\forall x \in X)[(h_k(x) = 1) \rightarrow (h_j(x) = 1)]$$

Forexample, according to the following graph, D1 is not more general than D2.



Whether it holds depends on the class, which the node belongs to before it is expanded, and the class which the node belongs to after it is expanded.

If the class of node is not clear before expanded, it can't be judged.

Assume $x = (\text{Sunny}, \text{Normal})$, $h_{D2}(x) = \text{Yes}$, but $h_{D1}(x) = \text{No}$. Because ID3 algorithm will choose the most common label.